VARYNX: Context-Dependent Factual Inconsistencies in Apertus

Track: Apertus Readiness — Track 1A **Event:** HackApertus 2026 **Team:** VARYNX — Lerato Mokgatla
Demo: Evaluation artefacts included in this repository **Finding:** `findings/varynx_001.json`

Summary

VARYNX investigates whether Apertus produces consistent, culturally and geographically appropriate factual answers when asked about South Africa's constitutional official languages. We evaluated responses across four combinations of geographic context and academic or research framing, using three recorded outputs per condition. The collected outputs show inconsistencies in enumeration and explanation, providing a candidate finding for further investigation rather than a confirmed model vulnerability.

Use of Apertus

- **Configured model identifier:** `swiss-ai/apertus-v1.5-8b`
- **Provider:** PublicAI hosted API-compatible endpoint.
- **Purpose:** Red-teaming and evaluation.
- **Execution:** Hosted inference through the VARYNX Flask application.
- **Verification limitation:** The configured identifier was recorded, but the returned checkpoint was not independently verified. Provider-level metadata was insufficient to establish the exact serving configuration.

Problem

Questions about a country's constitutional official languages are straightforward factual tasks, but incorrect answers can misrepresent linguistic communities and their legal recognition. This is particularly important in multilingual contexts where naming conventions, language varieties, historical changes and constitutional amendments must be distinguished carefully.

This investigation focuses on South Africa's official-language list and the recognition of South African Sign Language (SASL). The expected behaviour is a consistent, factually supported enumeration that treats Sepedi and Northern Sotho as names for the same language, does not count isiPedi and Sepedi as separate constitutional languages, and distinguishes an amendment's enactment from its commencement.

Primary-source reference for the commencement notice:
https://www.gov.za/documents/notices/constitution-eighteenth-amendment-act-commencement-act-26-jun-2026

Red-Teaming Method

Overall strategy

We compared responses across two contextual dimensions:

1. Geographic context: South Africa or the United Kingdom.
2. Prompt framing: research-oriented or academic.

The factual task remained focused on enumerating South Africa's constitutional official languages and explaining the relevant recognition timeline. The goal was to identify observable differences without assuming that any difference must be caused by a particular model mechanism.

Discovery process

Four prompt conditions were evaluated, with three recorded outputs per condition, for twelve outputs in total.

In the South Africa/research condition, responses listed Khoekhoe and omitted Sepedi while describing the recognition timeline inaccurately. In the South Africa/academic condition, responses listed isiPedi and Sepedi separately and omitted SASL. In the United Kingdom/research condition, the response included Sepedi but was truncated. In the United Kingdom/academic condition, the response included SASL.

These outputs demonstrate inconsistencies in enumeration, naming and historical explanation across the recorded conditions. The three captured outputs within each condition were identical. Provider-side caching or other serving behaviour cannot be ruled out, so the repetitions should not be treated as proof of independent stochastic replication. The results also do not establish that geographic context alone caused the differences.

Tooling

VARYNX uses a Flask application to support its evaluation workflow. The recorded evaluation artefact is `src/varynx-610b5e6c69.json`; the candidate finding is stored in `findings/varynx_001.json`.

A Docker-based run path is provided through the Track 1A Makefile. GitHub Actions is configured to verify the same `make run` command intended for judges. A successful CI run must be confirmed before clean-checkout execution is claimed as verified.

Findings Dataset

One candidate finding is documented: **Context-dependent factual inaccuracies in South African official-language enumeration.**

The evidence records conflicting lists and explanations concerning Khoekhoe, Sepedi, isiPedi and South African Sign Language. The finding contains example walkthroughs, reproduction steps, intended and observed behaviour, and explicit limitations.

VARYNX currently records this as an unverified candidate. The project does not claim independent reproduction against a verified serving checkpoint, nor does it establish a causal mechanism. The evidence supports further investigation of factual consistency and context sensitivity, not a definitive causal conclusion.

Lessons Learnt and Recommendations

- Re-test the prompts with provider caching disabled where possible, retaining request identifiers and response metadata.
- Verify the served model checkpoint and record provider, model revision, endpoint configuration, timestamps and generation settings.
- Repeat the evaluation with independent trials and document rate limits or truncation.
- Expand testing to additional South African languages, naming variants, constitutional references and historically changing facts.
- Compare outputs with authoritative sources and distinguish factual errors from incomplete or truncated answers.
- Preserve raw outputs and prompt configurations to support independent reproduction.

Reproducibility

The evaluation artefact records twelve outputs across four prompt conditions. The configured model identifier is `swiss-ai/apertus-v1.5-8b`, accessed through a hosted API-compatible endpoint. The recorded configuration includes temperature 0 and a maximum output setting of 350 tokens.

The provider-level information does not independently verify the returned checkpoint or exclude caching. The captured repetitions are not claimed to be statistically independent. These limitations should be addressed in a subsequent evaluation.

Run the project from the `track_1a` directory using `make run`. The command builds and launches the Docker application. Docker was unavailable in the local development PowerShell environment during report preparation; a passing GitHub Actions run is required to confirm execution from a clean checkout.

Submission commit: Record the final commit hash after changes have been reviewed and committed.

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References

1. South African Government, Constitution Eighteenth Amendment Act commencement notice: https://www.gov.za/documents/notices/constitution-eighteenth-amendment-act-commencement-act-26-jun-2026
2. Apertus model reference: https://huggingface.co/swiss-ai/Apertus-v1.5-8B
3. Evaluation export and candidate finding included with this submission.